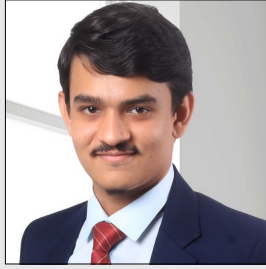


Dr. Parth Nayak

Data Scientist | AI Researcher | Astrophysicist



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🌐 par-nay

EDUCATION

🎓 **Ph.D. in Astrophysics** (Dr. rer. nat.)
LMU Munich | Dec 2025
🎓 **B.Sc.+M.Sc. in Physics** (dual degree)
IISER Bhopal | Jul 2020

SKILLS

ML / Deep Learning

Neural networks (CNNs, MLPs),
simulation-based inference (SBI),
generative models (normalizing flows,
diffusion models, autoencoders, GANs)

Other AI

Gaussian processes, Bayesian & Monte
Carlo methods, mixture models, transfer
learning, evolutionary algorithms

Programming

Python, LaTeX (proficient); HTML, CSS,
Javascript (intermediate); C++, Java
(basic)

Python packages

TensorFlow, Keras, Optuna, NumPy,
SciPy, Pandas, Matplotlib, H5Py,
AstroPy, HealPy, *among others*

Other IT

High-performance computing (HPC), Git
version control, LLM prompt
engineering, Markdown, Mac/MS Office,
Shell (bash, Z shell)

LANGUAGES

English (full working proficiency); Gujarati,
Hindi (native/bilingual); German (B1/
intermediate); French (beginner)

HIGHLIGHTS

- 5+ years of research experience in Data Science and AI algorithm development
- Authored 4 research papers on novel AI applications in astrophysics
- Delivered 10+ technical talks at international conferences and research institutes
- Received 5+ academic awards and fellowships

RELEVANT EXPERIENCE

Doctoral Candidate: Astrophysics & Data Science LMU Munich | Oct 2021 - Dec 2025
*Hightech Agenda Bayern chair of Astrophysics, Cosmology & AI + Munich Center for
Machine Learning (MCML)*

- Developed a ML inference framework “**LyaNNA**” [🌐 par-nay/lyanna] based on a novel 1D ResNet with Python/TensorFlow from scratch that **outperforms conventional methods** by >100% in inference precision for complex, realistic, noisy spectroscopic data (simulation-based inference)
- Developed “**SynTH**” [🌐 par-nay/synth] to model intergalactic absorption from massive cosmological **hydrodynamic simulations** and synthesize training data for *LyaNNA*
- Authored 3 research papers on AI for inference in astrophysics [✍Nayak+ 2024, 2025, Chang+ 2025]

Master's Thesis: Astrophysics & Data Science IISER Bhopal | Aug 2019 - Jul 2020

- Developed a Genetic Algorithm framework “**GA-ILC**” to optimize the reconstruction of cosmic microwave background (CMB) from contaminated observations
 - ▶ A novel, **scalable** evolutionary algorithm [🌐 par-nay/genepy] to extract faint cosmological signals from high-noise environments (foreground removal)
- Authored one research paper on evolutionary algorithms in astrophysics [✍Nayak & Saha 2022]

LEADERSHIP & SUPERVISION

- Co-supervisor of LMU Master's theses and labs Oct 2021 - Dec 2025
Designed and supervised two LMU Master's theses, and co-supervised 5+ computational and instrumental lab courses
- Co-organizer of “Astronomy on Tap” (AoT) Munich since Oct 2025
Coordinating the monthly outreach event to engage the general public with accessible scientific talks and pub-quizzes
- Co-organizer of “Meet the Speaker” @LMU Observatory Oct 2023 - May 2025
Revitalized the weekly series by relocating it to the historic 200-year old telescope dome, a unique atmosphere for students for informal networking with international experts
- Coordinator & Senior Advisor of IISERB Astronomy Club (IBAC) 2016 - 2020
Played a foundational role as the club's second coordinator, establishing enduring traditions and growing the community into one of India's most popular student astronomy clubs
- Physics Department Representative Aug 2017 - Jul 2018
Unanimously elected to serve as the primary liaison between faculty and a class of 60+ physics major students

SELECTED AWARDS & FELLOWSHIPS

- DAAD Research Grant — Ph.D. scholarship Germany | Oct 2021 - Dec 2025
- Proficiency Medal in Physics — IISER Bhopal India | Jul 2020
- DAAD-WISE Scholarship Germany | May - Jul 2019
- INSPIRE Scholarship for Higher Education India | Aug 2015 - Jul 2020